8.3. Setting up Super Mario Bros.

Together, the forward, inverse, and encoder models form the *intrinsic curiosity module* (ICM), which we will discuss in detail later in this chapter. The components of the ICM function together for the sole purpose of generating an intrinsic reward that drives curiosity in the agent. The ICM generates a new intrinsic reward signal based on information from the environment, so it is independent of how the agent model is implemented. The ICM can be used for any type of environment, but it will be most useful in sparse reward environments.

We could use whatever agent model implementation we want, such as a distributed actor-critic model (covered in chapter 5). In this chapter we will use a Q-learning model to keep things simple and focus on implementing the ICM. We will use Super Mario Bros. as our testbed.

Super Mario Bros. does not really suffer from the sparse reward problem. The particular environment implementation we will use provides a reward in part based on forward progress through the game, so positive rewards are almost continuously provided. However, Super Mario Bros. is still a great choice to test the ICM because we can choose to "turn off" the extrinsic (environment-provided) reward signal; we can see how well the agent explores the environment just based on curiosity, and we can see how well correlated the extrinsic and intrinsic rewards are.

The implementation of Super Mario Bros. we will use has 12 discrete actions that can be taken at each time step, including a NO-OP (no-operation) action. Table 8.1 lists all the actions.

Table 8.1. Actions in Super Mario Bros.

Index	Action
0	NO-OP / Do nothing
1	Right
2	Right + Jump
3	Right + Run
4	Right + Jump + Run